

Product Reference Manual (V1.0)

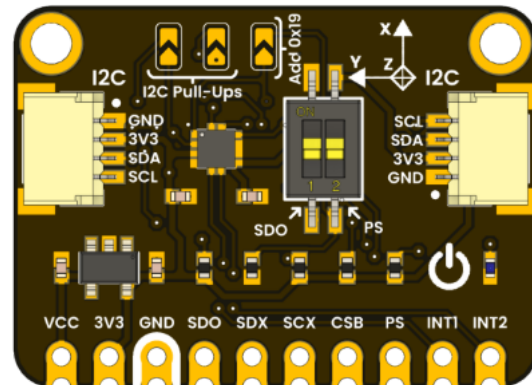
Description

This module features the **BMA255**, a high-performance **3-axis digital accelerometer** engineered for precise motion sensing and orientation detection. It measures acceleration along the X, Y, and Z axes with excellent accuracy, making it suitable for a wide range of applications including wearable devices, robotics, and mobile systems. The sensor's **I²C interface** ensures easy communication with most microcontrollers, while its optional **SPI interface** allows for higher data throughput in demanding applications.

The module is built on the **DevLab Factor** standard, offering a compact and well-labeled layout with dual I²C connectors, address selection switch, and configurable pull-up resistors for simplified integration. It supports programmable measurement ranges ($\pm 2g$ to $\pm 16g$), low-pass filters, and multiple motion-triggered interruptions such as tap, double-tap, orientation, and free-fall detection. Designed for **low power consumption**, it is ideal for battery-powered devices and embedded projects that require reliable motion tracking and intelligent event detection.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab module



Key Features

- **Ultra-compact form factor:** 2 mm × 2 mm × 0.95 mm LGA package
- **Digital Interfaces:** I²C and SPI (3-wire / 4-wire compatible)
- **Supply Voltage:**
 - VDD: 1.62 V – 3.6 V
 - VDDIO: 1.2 V – 3.6 V
- **Measurement Ranges:** $\pm 2 g$ / $\pm 4 g$ / $\pm 8 g$ / $\pm 16 g$
- **Low-Pass Filter Bandwidths:** Selectable from 1 kHz down to < 8 Hz
- **Built-in FIFO:** 32-frame deep data buffer for efficient sampling
- **Interrupt Features:**
 - New data
 - Any-motion (slope) detection
 - Tap / Double-tap recognition
 - Orientation & flat position detection
 - Low-g / High-g events
 - Inactivity detection
- **Compact DevLab Form Factor:** Includes a **QWIC-compatible JST 1.0 mm connector** for quick and secure I²C expansion.

Software Support

- **Arduino IDE**
- **Micro Python**

Applications

- Display orientation and auto-rotation
- Tap / double-tap gesture control
- Step counting and pedometer functions
- Free-fall and drop detection
- E-compass tilt compensation
- Power-management trigger for handhelds
- Motion-activated logging systems

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1 The Board

1.1 Accessories

- 1×10-pin 2.54 mm male header
- 1x QWIIIC 4-pin JST 1.0 mm Harness

2 Ratings

2.1 Recommended Operating Conditions

Parameter	Description	Min	Typ	Max	Unit
V_{in}	Input voltage to power on the module	2.5	-	5.5	V
V_{dd}	Supply Voltage for sensor	1.62	-	3.6	V
V_{ddio}	Supply Voltage for I/O	1.2	2.4	3.6	V
I_{dd}	Total Supply Current	6.5	-	130	uA
I_{3v3(1)}	Maximum current of 3V3 regulator	-	-	600	mA
V_{ih}	Input high level voltage	0.7 V _{ddio}	-	-	V
V_{il}	Input low level voltage	-	-	0.3 V _{ddio}	V
V_{ol}	Output low level voltage	-	-	0.2 V _{ddio}	V
V_{oh}	Output high level voltage	0.8 V _{ddio}	-	-	V
S_{xg(2)}	Sensitivity	128	-	1024	LSB/g

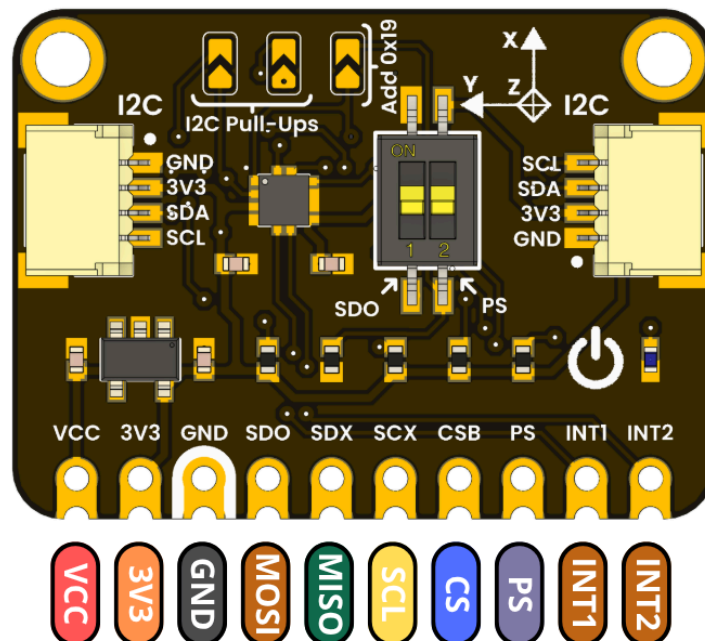
3 Functional Overview

3.1 General Description

The **DevLab BMA255 Module** integrates the **BMA255**, an ultra-small, low-power, triaxial digital acceleration sensor intended for consumer and embedded electronics applications. The BMA255 provides **12-bit digital resolution**, enabling low-noise and high-accuracy measurement of acceleration along three orthogonal axes. This capability allows reliable detection of tilt, motion, shock, and vibration, supporting a wide range of motion-aware system functions.

The sensor is housed in a compact **2 mm × 2 mm LGA package**, making it particularly suitable for space-constrained designs such as smartphones, handheld devices, computer peripherals, human-machine interfaces, virtual-reality systems, and gaming controllers. Its small footprint and low power consumption facilitate integration into modern portable and battery-powered devices.

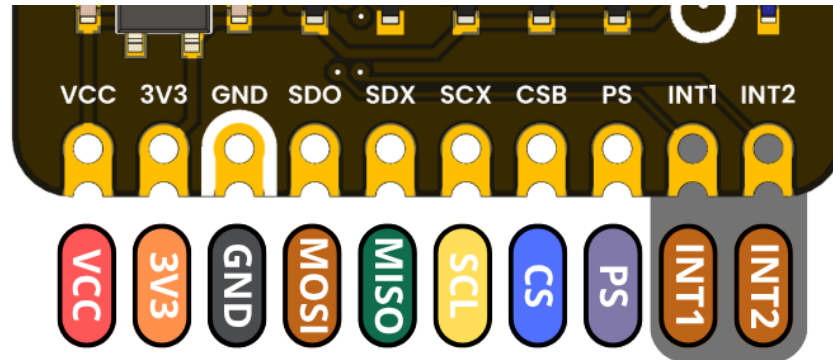
The module is designed in accordance with the **DevLab Form Factor**, which prioritizes simplicity, modularity, and interoperability. The hardware layout includes **dual I²C connectors** for easy bus expansion, an **address selection mechanism** to support multiple devices on the same bus, and **configurable pull-up resistors** to accommodate different system requirements. This standardized design enables rapid integration into embedded platforms and ensures seamless compatibility with other DevLab modules.



Overall, the DevLab BMA255 Module provides a compact, efficient, and flexible motion-sensing solution, suitable for rapid prototyping, educational platforms, and production-ready embedded systems requiring reliable acceleration measurement and intelligent motion detection.

3.2 Interrupt Interface (INT1 / INT2)

The **BMA255** provides two independent, programmable interrupt outputs that allow the sensor to notify the host system of motion-related events without continuous polling. These interrupt pins are controlled by the internal interrupt engine and configured through software-accessible registers.



Interrupt Pins Description

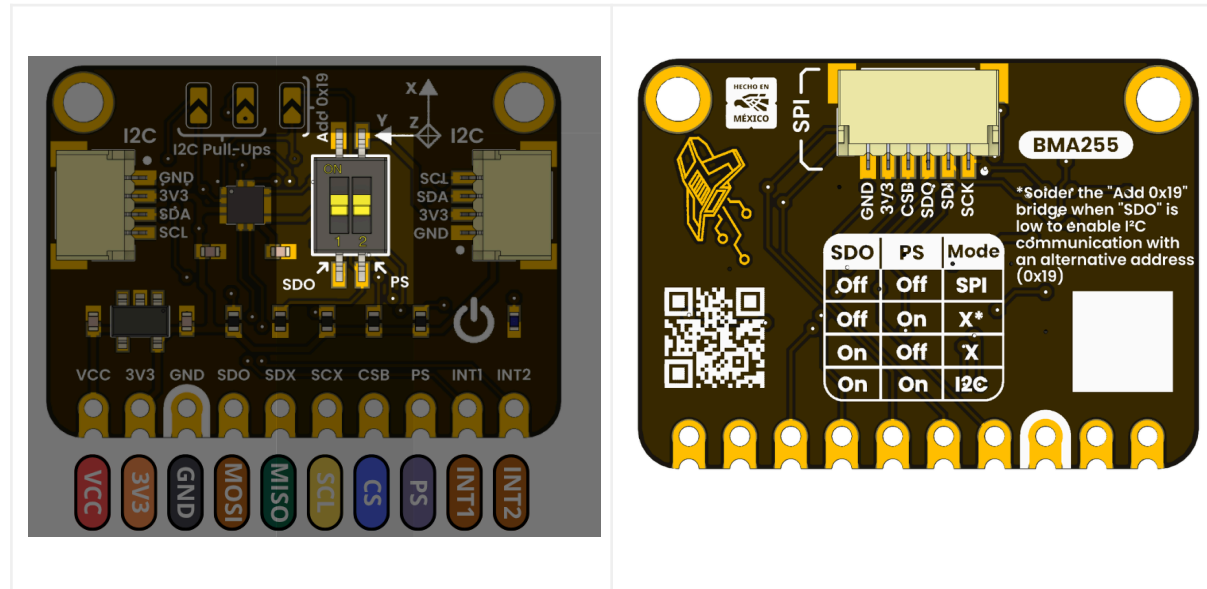
Pin	Signal Type	Function	Configuration and Operating Conditions
INT2	Output	Interrupt output 2	Fully configurable interrupt pin. Can signal motion, tap, orientation change, free-fall, no-motion, or other events as defined by internal registers.
INT1	Output	Interrupt output 1	Primary interrupt output for data-ready, motion detection, free-fall, and system wake-up events. Interrupt sources are software-mapped.

Configuration notes:

- Interrupt sources can be freely mapped to **INT1**, **INT2**, or both.
- Thresholds, hysteresis, timing, and polarity are programmable.
- Enables low-power operation by allowing the host MCU to remain in sleep mode until an event occurs.

3.3 Communication Mode Selection (PS, I²C, SPI)

The communication protocol of the BMA255 is selected using the **PS (Protocol Select)** pin. This allows the same hardware design to support either **I²C** or **SPI**, depending on system requirements.



Protocol Selection Pin

Pin	Signal Type	Function	Description
PS	Input	Protocol select	LOW = I²C mode, HIGH = SPI mode. Must be held at a valid logic level during operation.

Communication Mode and Address Configuration Table

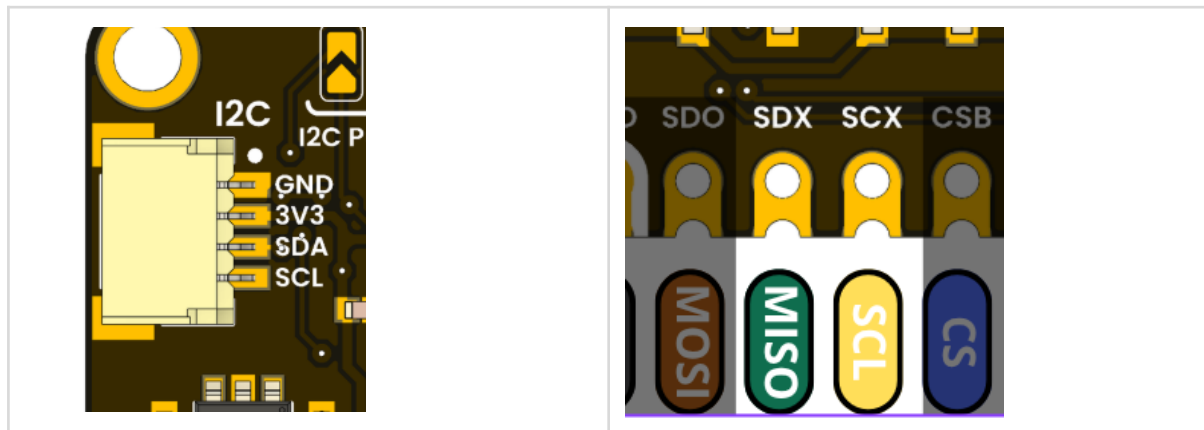
(According to DevLab BMA255 Module Hardware Layout)

SDO	PS	Selected Mode	Description / Operating Condition
OFF	OFF	SPI	Default SPI configuration. SDO disabled. SPI signals (SCK, MOSI, MISO, CS) active.
OFF	ON	X*	Reserved / not recommended configuration. Not a valid standard operating mode.
ON	OFF	X	Invalid configuration. Do not use.

ON	ON	I ² C	I ² C mode enabled. SDO acts as address select when bridged.
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3.4 I²C Communication Interface

When **PS = LOW**, the BMA255 operates in **I²C mode**, providing a simple two-wire interface suitable for low-pin-count designs and shared sensor buses.



I²C Pin Usage

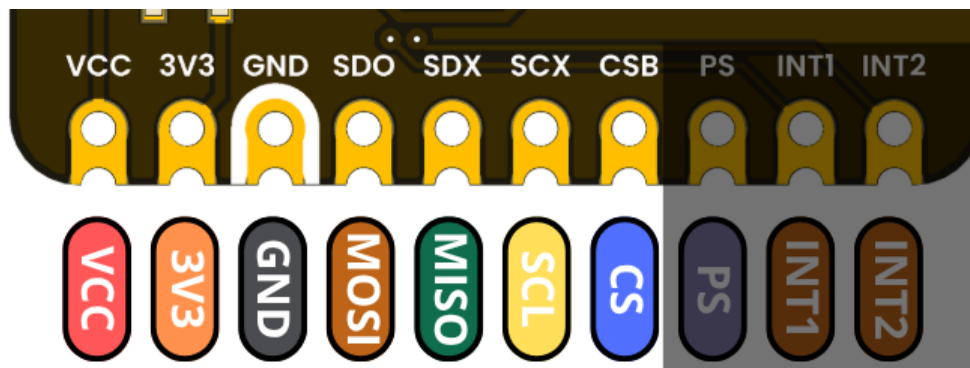
Pin	I ² C Role	Description
MISO (SDO)	SDA	I ² C data line (bidirectional). Functions as SDA in I ² C mode.
SCL	SCL	I ² C clock line.
CS	—	Must be tied HIGH in I ² C mode.
MOSI	—	Not used in I ² C mode.

I²C configuration notes:

- Supports standard I²C operation with register-based access.
- Address selection is handled externally (DevLab address switch).
- Suitable for low-power and multi-device bus configurations.

3.5 SPI Communication Interface

When **PS = HIGH**, the BMA255 operates in **SPI mode**, enabling higher data throughput and deterministic timing.



SPI Pin Usage

Pin	SPI Role	Description
SCL	SCK	SPI serial clock.
MOSI	SDI	Master Out / Slave In (data input to BMA255).
MISO	SDO	Master In / Slave Out (data output from BMA255).
CS	CS	Chip Select, active LOW . Enables SPI communication.

SPI configuration notes:

- Supports 3-wire or 4-wire SPI operation.
- CS must be asserted LOW for each transaction.
- Provides higher bandwidth compared to I²C.

3.6 Power Supply and Regulation (VCC / 3V3)

The DevLab BMA255 module includes an onboard **AP2112K-3.3V** regulator, clearly separating unregulated input power from the regulated operating voltage.

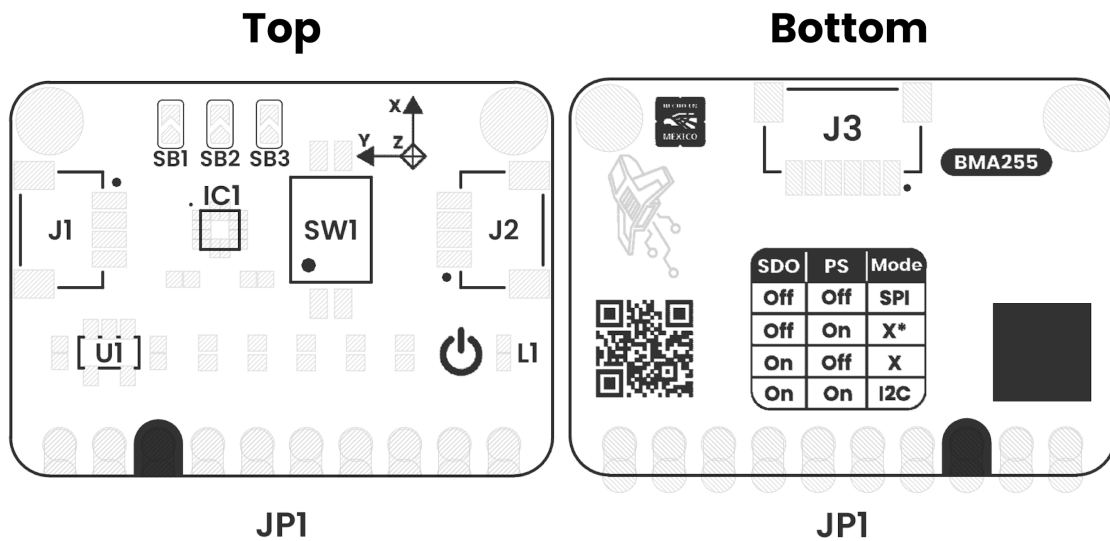
3.7 Power Pins Description

Pin	Role	Description and Conditions
VCC	VIN (unregulated input)	Input to AP2112K-3.3V regulator. Accepts 2.5 V to 5 V (USB, battery, system rail).
3V3	Regulated output	+3.3 V regulated rail. Supplies sensor I/O and logic. May be used as an external 3.3 V input only if VCC is not connected .
GND	Ground	Common ground reference. Must be connected to system ground.

Power design rules:

- Do not connect external 3.3 V to 3V3 while also powering VCC.
- Choose VCC for unregulated supplies and 3.3V for regulated systems.
- Ensures safe operation and prevents regulator backfeeding.

3.8 Board Topology



Views of Board Topology

Views of Topology

Table 3.2.1 - Components Overview

Ref.	Description
IC1	BMA255
L1	Power On LED
U1	AP2112K 3V3 Regulator
JP1	2.54 mm Castellated Holes
J1	QWIIC Connector (JST 1 mm pitch) for I2C
J2	QWIIC Connector (JST 1 mm pitch) for I2C
J3	JST 1 mm pitch for SPI
SW1	Dip Switch for Mode Selection
SB1	Solder Bridge for I2C Pull-Ups
SB2	Solder Bridge for I2C Pull-Ups
SB3	Solder Bridge for I2C Address

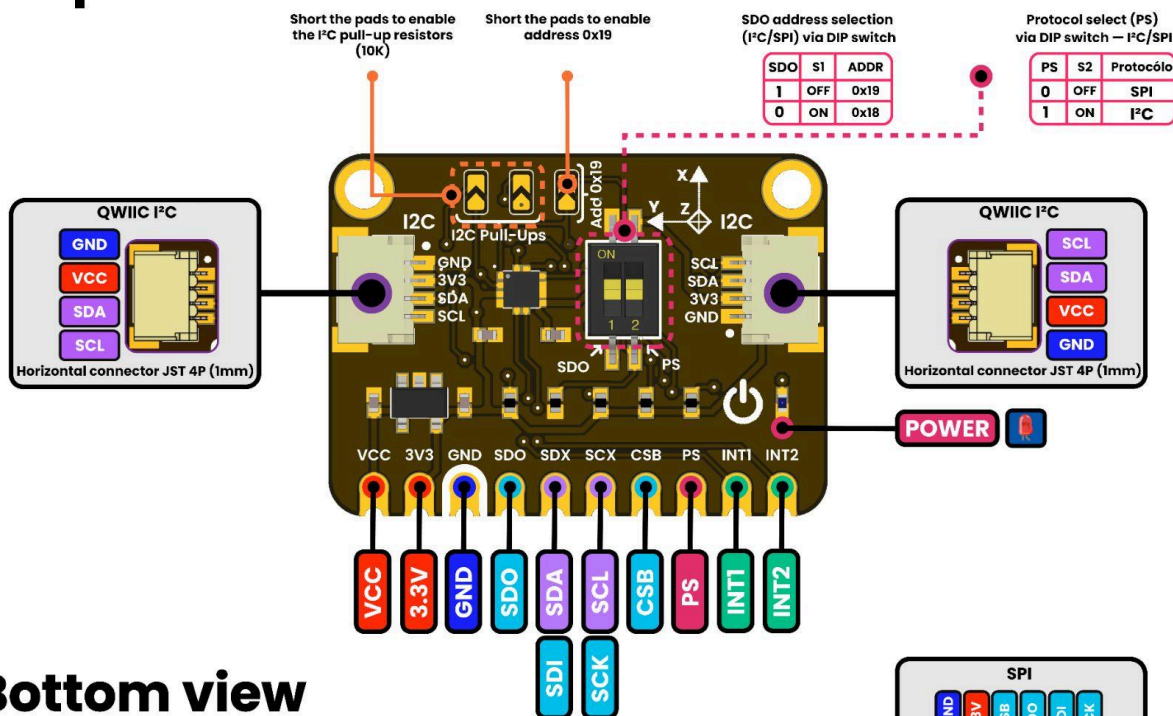
4 Connectors & Pinouts

4.1 General Pinout

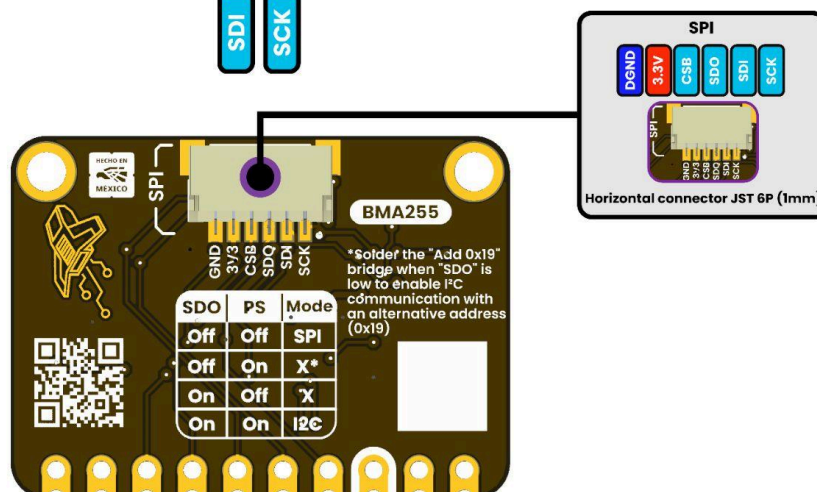
PINOUT

DevLab: I2C BMA255 Ultra-Low Power Accelerometer Sensor

Top view



Bottom view



General Pinout

4.2 Pinout General Description

Pin Label	Function	Notes
INT2	Interrupt output 2	Configurable interrupt pin (e.g. motion, tap, orientation). Can be mapped via registers.
INT1	Interrupt output 1	Main interrupt output for events such as data ready, motion, free-fall, etc.
PS	Protocol select	Selects communication mode: LOW = I²C , HIGH = SPI
CS	Chip Select (SPI)	Active LOW. Used only in SPI mode. Tie HIGH in I ² C mode.
SCL	Serial Clock	I ² C clock (SCL) or SPI clock (SCK) depending on PS pin
MISO	Master In Slave Out	SPI data output from BMA255 to master (SDO)
MOSI	Master Out Slave In	SPI data input to BMA255 from master (SDI)
GND	Ground	Ground reference
3V3	+3.3 V regulated rail	Digital I/O voltage (typically 3.3 V)
VCC	VIN (unregulated input)	Input to AP2112K-3.3V regulator. Feed 2.5–5 V if needed.

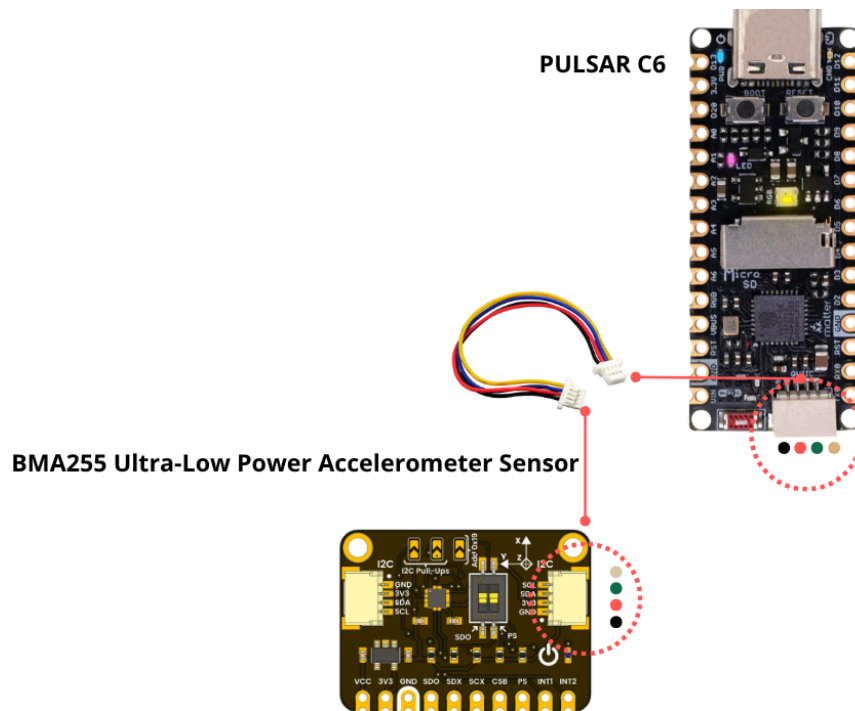
Important Notes

- In I²C mode, tie CS = HIGH and PS = LOW.
- In SPI mode, set PS = HIGH and use CS actively.
- INT1/INT2 are fully configurable and can be routed to multiple internal events.
- Ensure VCC and 3V3 meet the voltage requirements of your host MCU.

5 Board Operation

5.1 Getting Started with arduino IDE

This section describes the initial setup required to interface the **BMA250/BMA255 ultra-low power accelerometer** with the **PULSAR C6 (ESP32-C6)** using the Arduino IDE over the I²C protocol. The objective is to validate communication, initialize the sensor, and obtain real-time acceleration data.



Hardware Configuration

The module supports standard I²C communication and can be connected using a 4-pin JST (Qwiic/STEMMA QT) cable or direct pin wiring.

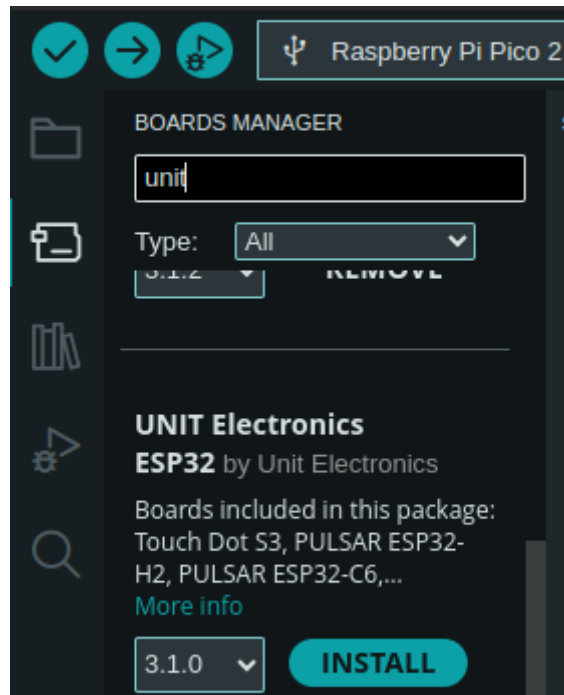
Electrical Connections:

Signal	Connection	Description
VCC	3.3V	Power supply (do not use 5V)
GND	GND	Common ground
SDA	GPIO6	I ² C data line
SCL	GPIO7	I ² C clock line
SDO	GND / VCC	I ² C address selection (0x18 / 0x19)

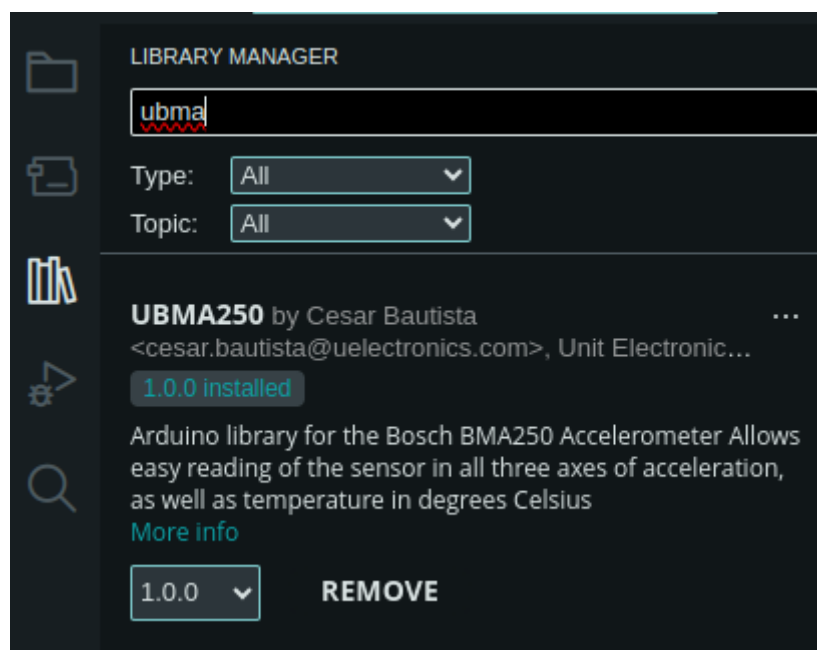
The ESP32-C6 allows flexible I²C pin assignment; however, GPIO6 (SDA) and GPIO7 (SCL) are recommended for consistency within the DevLab ecosystem.

Software Setup (Arduino IDE)

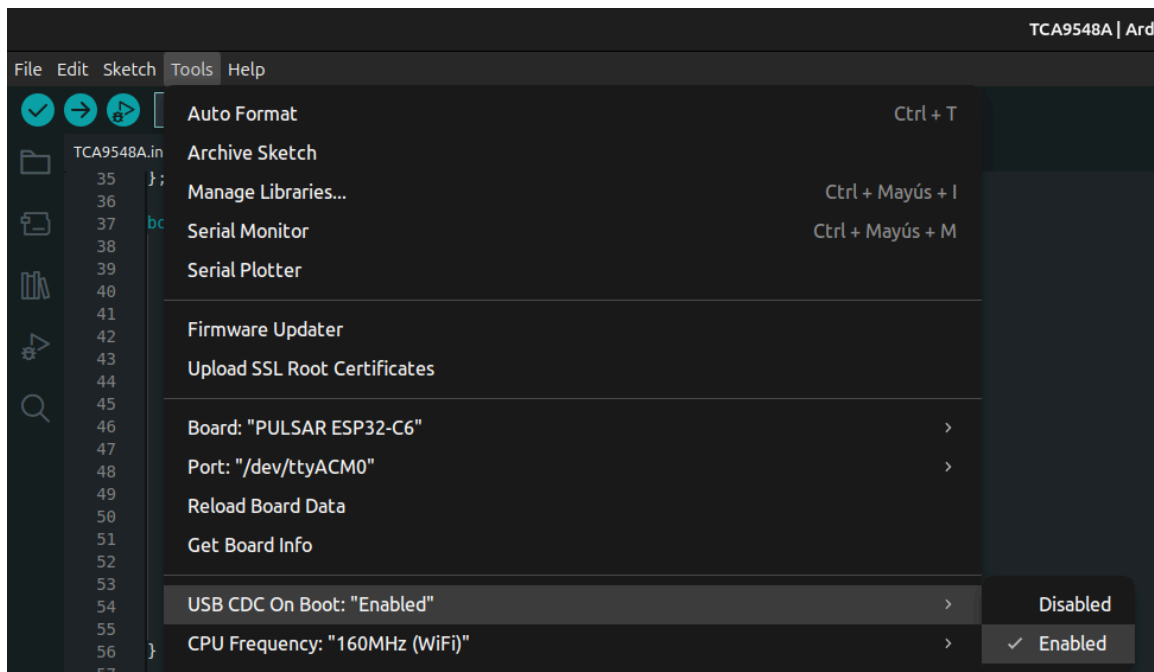
1. Install the **UNIT Electronics ESP32**



- 2.
3. Select the correct board **PULSAR ESP32-C6** (ESP32-C6).
4. Install the **BMA250 library** provided by UNIT Electronics.



5. Connect the board via USB-C and select the appropriate COM port.



Initialization and Communication Flow

The firmware performs the following sequence:

- Initializes Serial communication for debugging.
- Configures the I²C bus with custom GPIO pins.
- Attempts to initialize the BMA250 sensor with:
 - Range: $\pm 2g$
 - Update rate: 64 ms
- If initialization fails:
 - Performs an I²C scan to detect connected devices.
- Continuously reads acceleration (X, Y, Z) and temperature data.
- Outputs values to the Serial Monitor.

Arduino Example Code

```
#include <Wire.h>
#include "BMA250.h"

#define SDA_PIN 6
#define SCL_PIN 7

UBMA250 accel_sensor;
int x, y, z;
double temp;

void setup() {
    Serial.begin(115200);
```

```
while (!Serial && millis() < 5000) delay(10);

Wire.begin(SDA_PIN, SCL_PIN);

int result = accel_sensor.begin(BMA250_range_2g,
BMA250_update_time_64ms);

if (result != 0) {
  Serial.println("BMA250 initialization failed. Scanning I2C...");
  for (byte addr = 1; addr < 127; addr++) {
    Wire.beginTransmission(addr);
    if (Wire.endTransmission() == 0) {
      Serial.print("Device at 0x");
      Serial.println(addr, HEX);
    }
  }
}

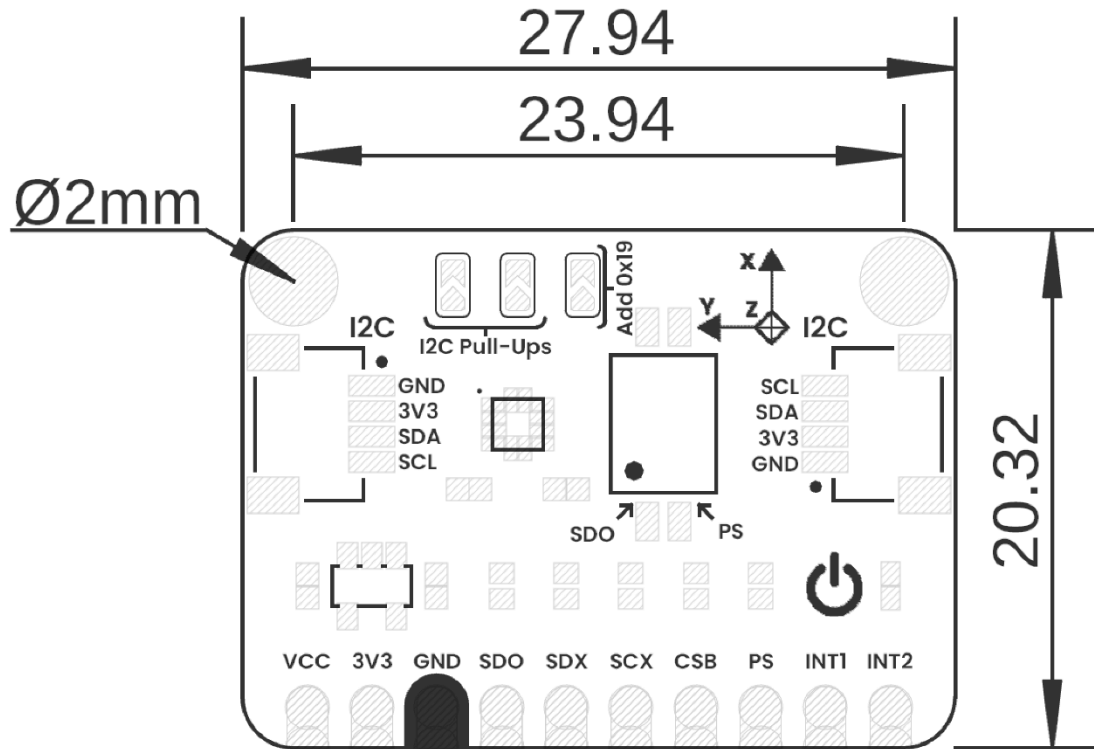
void loop() {
  accel_sensor.read();

  x = accel_sensor.X;
  y = accel_sensor.Y;
  z = accel_sensor.Z;
  temp = ((accel_sensor.rawTemp * 0.5) + 24.0);

  if (x == -1 && y == -1 && z == -1) {
    Serial.println("BMA250 not detected");
  } else {
    Serial.print("X="); Serial.print(x);
    Serial.print(" Y="); Serial.print(y);
    Serial.print(" Z="); Serial.print(z);
    Serial.print(" Temp="); Serial.println(temp);
  }

  delay(250);
}
```

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

8 Reference Documentation

Ref	Link
Documentation	https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bma255_ultra_low_power_accelerometer_sensor
Wiki	https://wiki.uelectronics.com/wiki/devlab_i2c_bma255_ultra_low_power_accelerometer_sensor
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(https://github.com/UNIT-Electronics-MX/unit_devlab_i2c_bma255_ultra_low_power_accelerometer_sensor/blob/main/hardware/unit_sch_v_1_0_0_ue0109_bma250_accelerometer_module.pdf.)

